

01_qc

Roger Calvas

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QC Report- Nanopore Sequencing

Sample: T1

Pre-Filtration QC

Statistics

General summary:

Mean read length:	598.9
Mean read quality:	16.3
Median read length:	321.0
Median read quality:	18.7
Number of reads:	8,634.0
Read length N50:	978.0
STDEV read length:	965.8
Total bases:	5,170,693.0

Number, percentage and megabases of reads above quality cutoffs

>Q10:	8528	(98.8%)	5.1Mb
>Q15:	6287	(72.8%)	3.9Mb
>Q20:	3673	(42.5%)	2.3Mb
>Q25:	1788	(20.7%)	0.8Mb
>Q30:	834	(9.7%)	0.3Mb

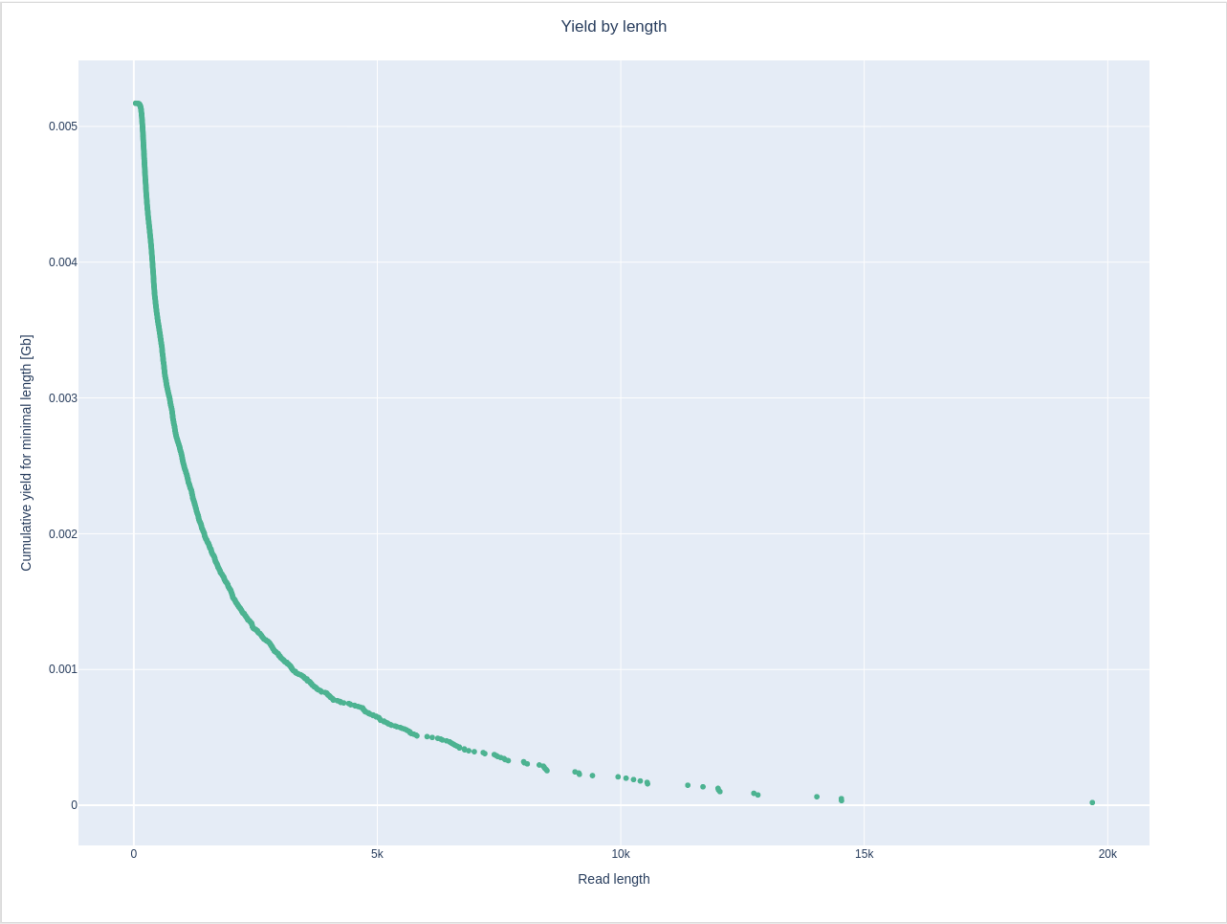
Top 5 highest mean basecall quality scores and their read lengths

1:	45.0	(149)
2:	43.8	(93)
3:	43.7	(113)
4:	43.6	(134)
5:	43.6	(153)

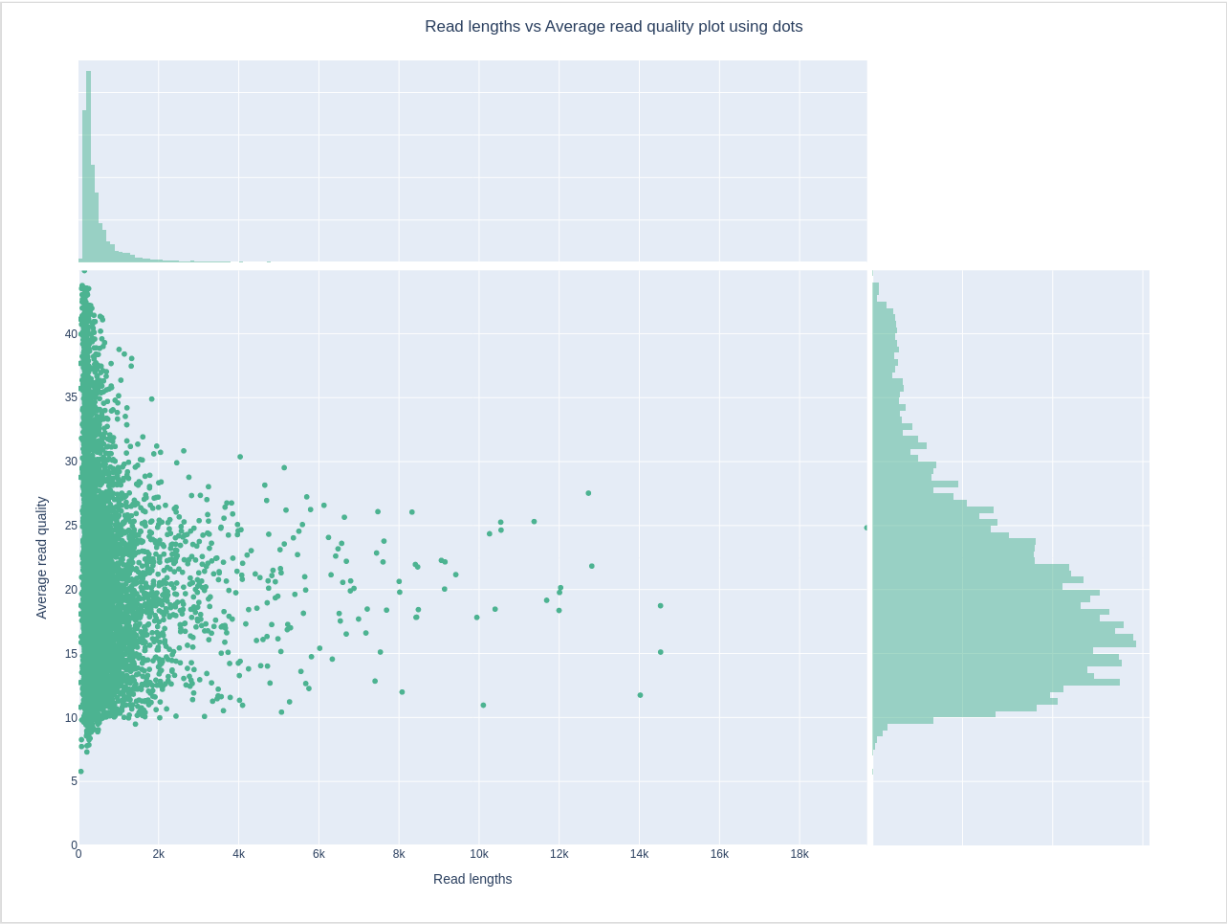
Top 5 longest reads and their mean basecall quality score

1:	19684	(24.8)
2:	14534	(15.1)
3:	14530	(18.7)
4:	14026	(11.8)
5:	12816	(21.8)

Yield by Length



Length vs Quality



Post-Filtration QC

Statistics

General summary:

Mean read length:	2,142.1
Mean read quality:	16.9
Median read length:	1,481.5
Median read quality:	19.3
Number of reads:	1,252.0
Read length N50:	2,419.0
STDEV read length:	1,858.1
Total bases:	2,681,925.0

Number, percentage and megabases of reads above quality cutoffs

>Q10:	1249	(99.8%)	2.7Mb
>Q15:	975	(77.9%)	2.2Mb
>Q20:	553	(44.2%)	1.2Mb
>Q25:	171	(13.7%)	0.3Mb
>Q30:	28	(2.2%)	0.0Mb

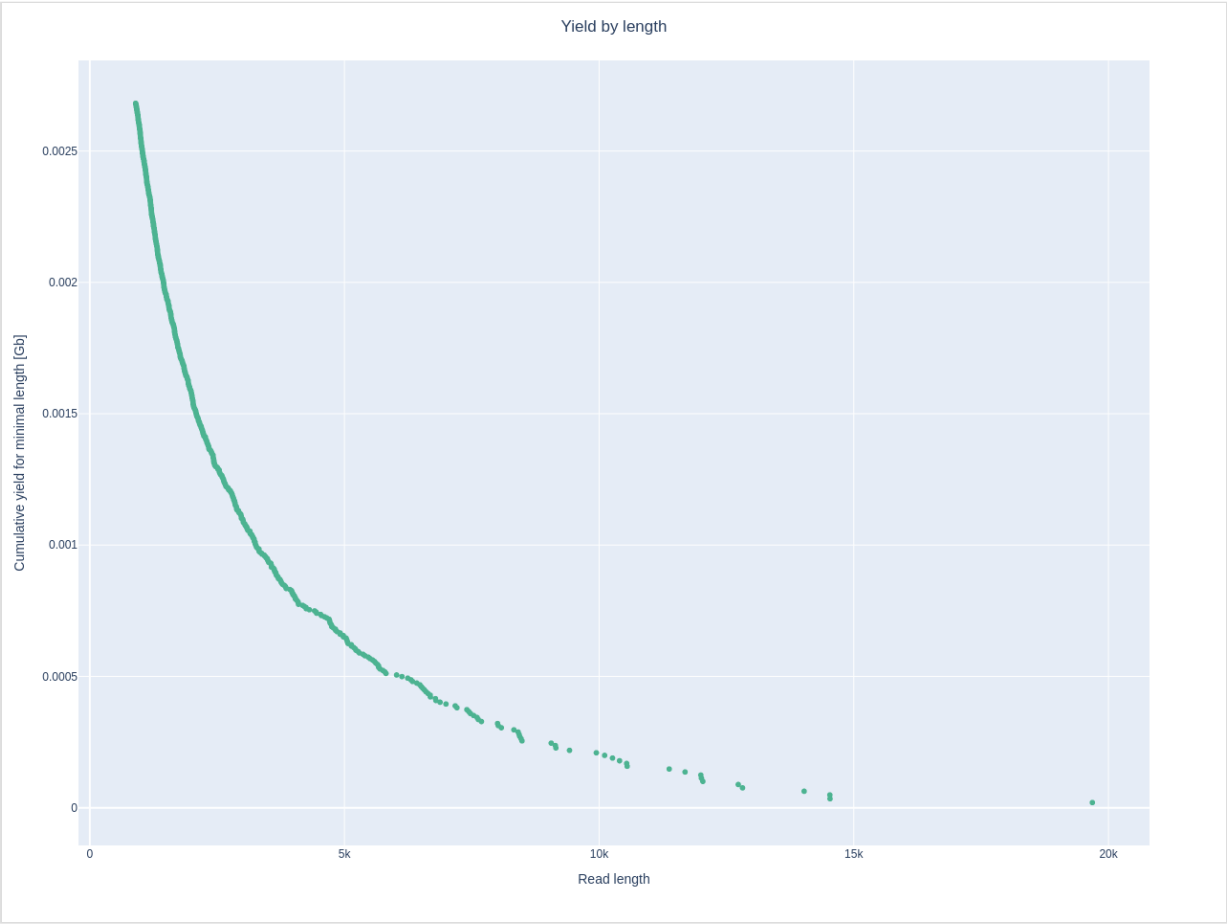
Top 5 highest mean basecall quality scores and their read lengths

1:	38.8	(1014)
2:	38.4	(1145)
3:	38.1	(1327)
4:	37.5	(1316)
5:	36.4	(1059)

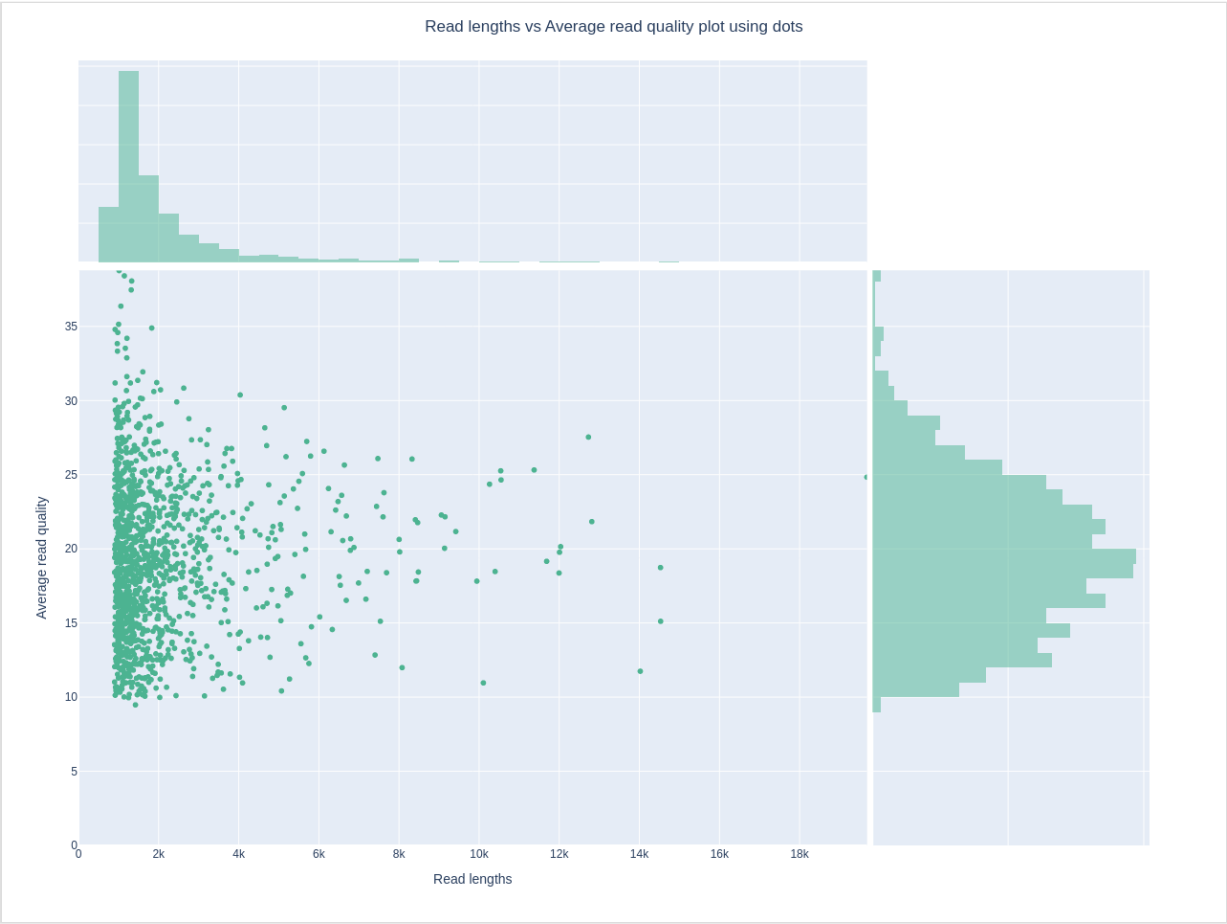
Top 5 longest reads and their mean basecall quality score

1:	19684	(24.8)
2:	14534	(15.1)
3:	14530	(18.7)
4:	14026	(11.8)
5:	12816	(21.8)

Yield by Length



Length vs Quality



Sample: M1

Pre-Filtration QC

Statistics

General summary:

Mean read length: 914.3

Mean read quality: 16.5

Median read length: 409.0

Median read quality: 19.1

Number of reads: 90,663.0

Read length N50: 1,943.0

STDEV read length: 1,512.4

Total bases: 82,890,408.0

Number, percentage and megabases of reads above quality cutoffs

>Q10: 89580 (98.8%) 82.5Mb

>Q15: 67329 (74.3%) 62.9Mb

>Q20: 40454 (44.6%) 35.3Mb

>Q25: 18806 (20.7%) 11.7Mb

>Q30: 8351 (9.2%) 3.1Mb

Top 5 highest mean basecall quality scores and their read lengths

1: 45.6 (141)

2: 45.4 (228)

3: 45.1 (162)

4: 44.9 (153)

5: 44.7 (163)

Top 5 longest reads and their mean basecall quality score

1: 40666 (12.4)

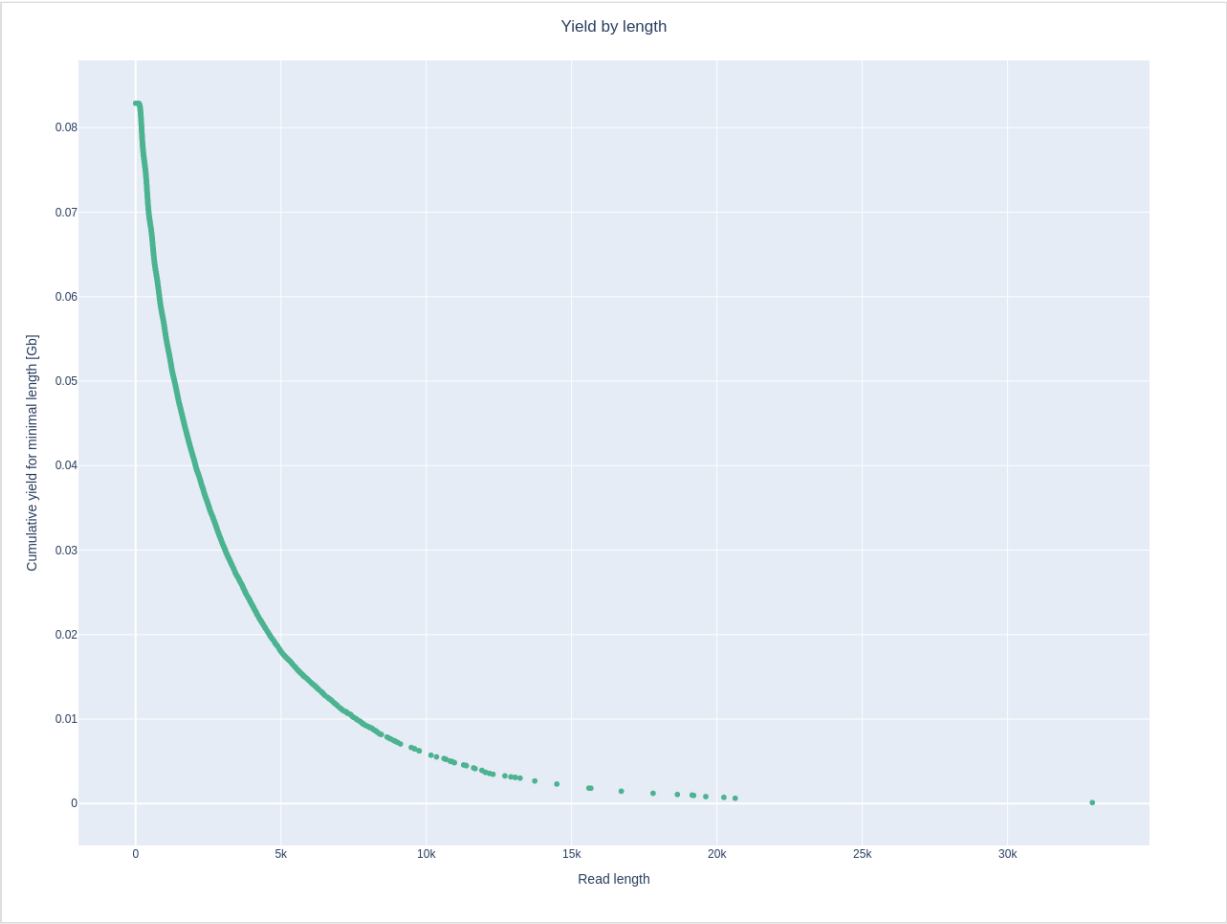
2: 40563 (15.9)

3: 32913 (17.0)

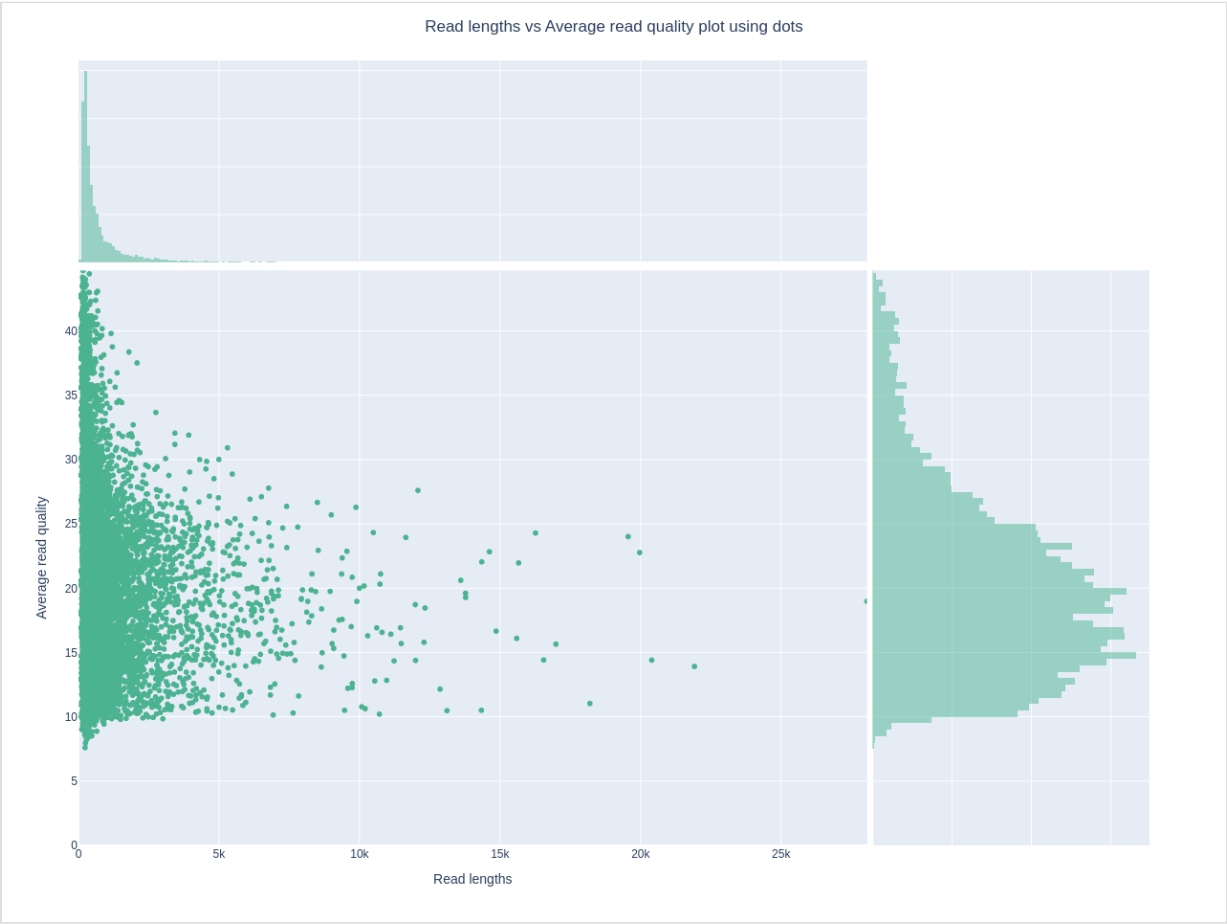
4: 31255 (18.2)

5: 28296 (13.0)

Yield by Length



Length vs Quality



Post-Filtration QC

Statistics

General summary:

Mean read length:	2,552.1
Mean read quality:	16.7
Median read length:	1,742.0
Median read quality:	19.0
Number of reads:	22,794.0
Read length N50:	3,197.0
STDEV read length:	2,323.7
Total bases:	58,172,104.0

Number, percentage and megabases of reads above quality cutoffs

>Q10:	22731	(99.7%)	58.0Mb
>Q15:	17455	(76.6%)	44.7Mb
>Q20:	9777	(42.9%)	24.5Mb
>Q25:	3045	(13.4%)	6.6Mb
>Q30:	543	(2.4%)	0.9Mb

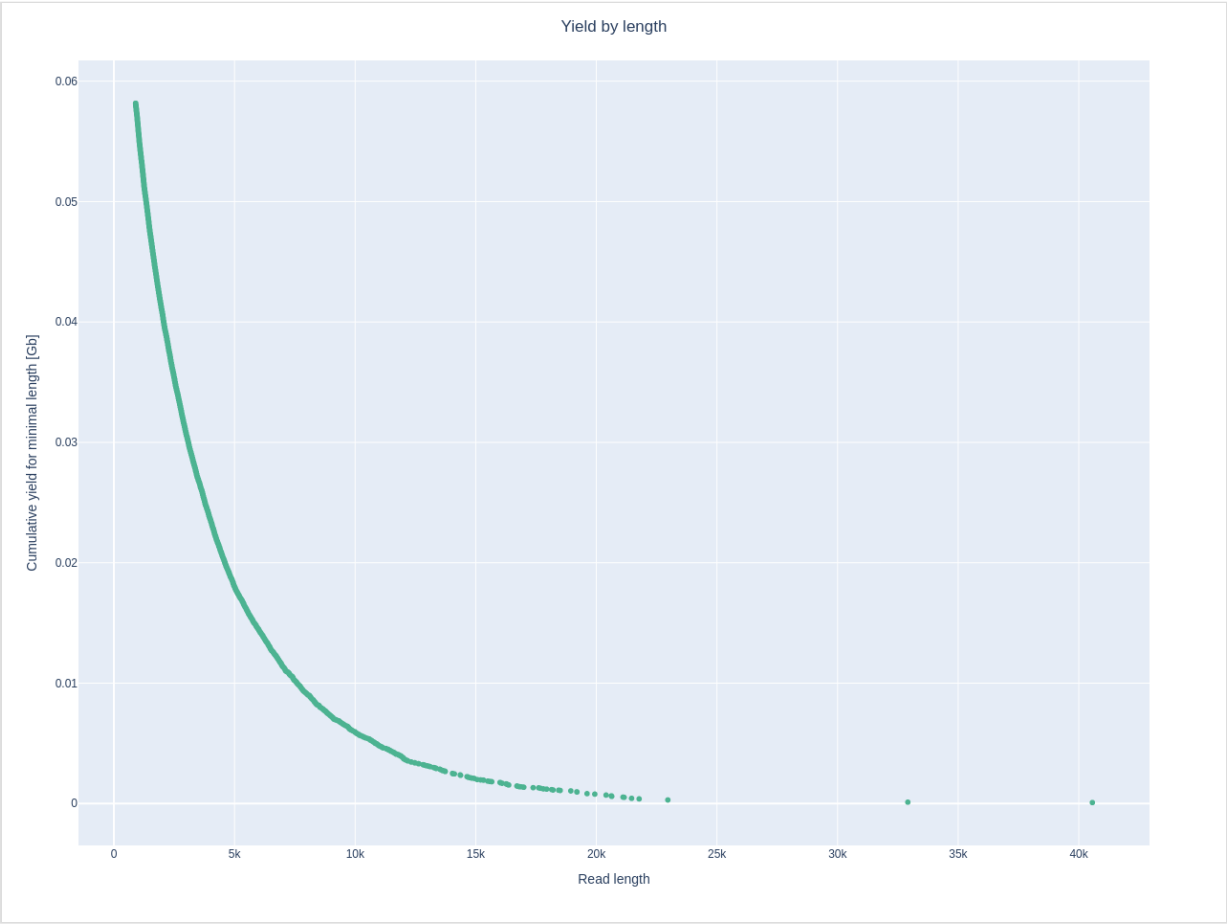
Top 5 highest mean basecall quality scores and their read lengths

1:	42.7	(932)
2:	41.9	(1054)
3:	41.8	(1834)
4:	40.1	(1189)
5:	39.8	(927)

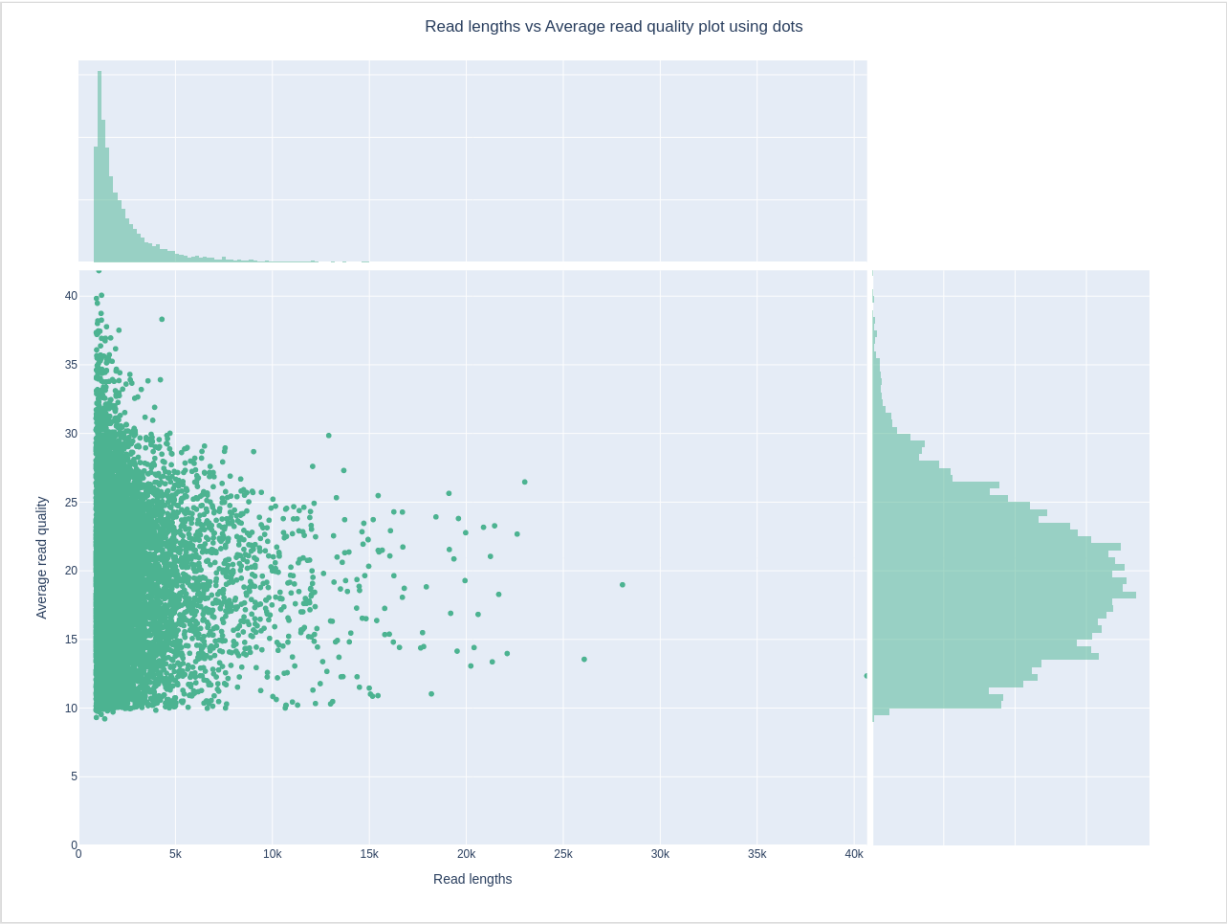
Top 5 longest reads and their mean basecall quality score

1:	40666	(12.4)
2:	40563	(15.9)
3:	32913	(17.0)
4:	31255	(18.2)
5:	28296	(13.0)

Yield by Length



Length vs Quality



Sample: A1

Pre-Filtration QC

Statistics

General summary:

Mean read length: 932.4

Mean read quality: 16.4

Median read length: 417.0

Median read quality: 19.0

Number of reads: 96,525.0

Read length N50: 1,965.0

STDEV read length: 1,532.0

Total bases: 89,999,785.0

Number, percentage and megabases of reads above quality cutoffs

>Q10: 95330 (98.8%) 89.6Mb

>Q15: 70991 (73.5%) 69.1Mb

>Q20: 42499 (44.0%) 38.5Mb

>Q25: 19804 (20.5%) 12.5Mb

>Q30: 8940 (9.3%) 3.3Mb

Top 5 highest mean basecall quality scores and their read lengths

1: 44.8 (218)

2: 44.4 (150)

3: 44.4 (181)

4: 44.3 (232)

5: 44.3 (152)

Top 5 longest reads and their mean basecall quality score

1: 49492 (14.3)

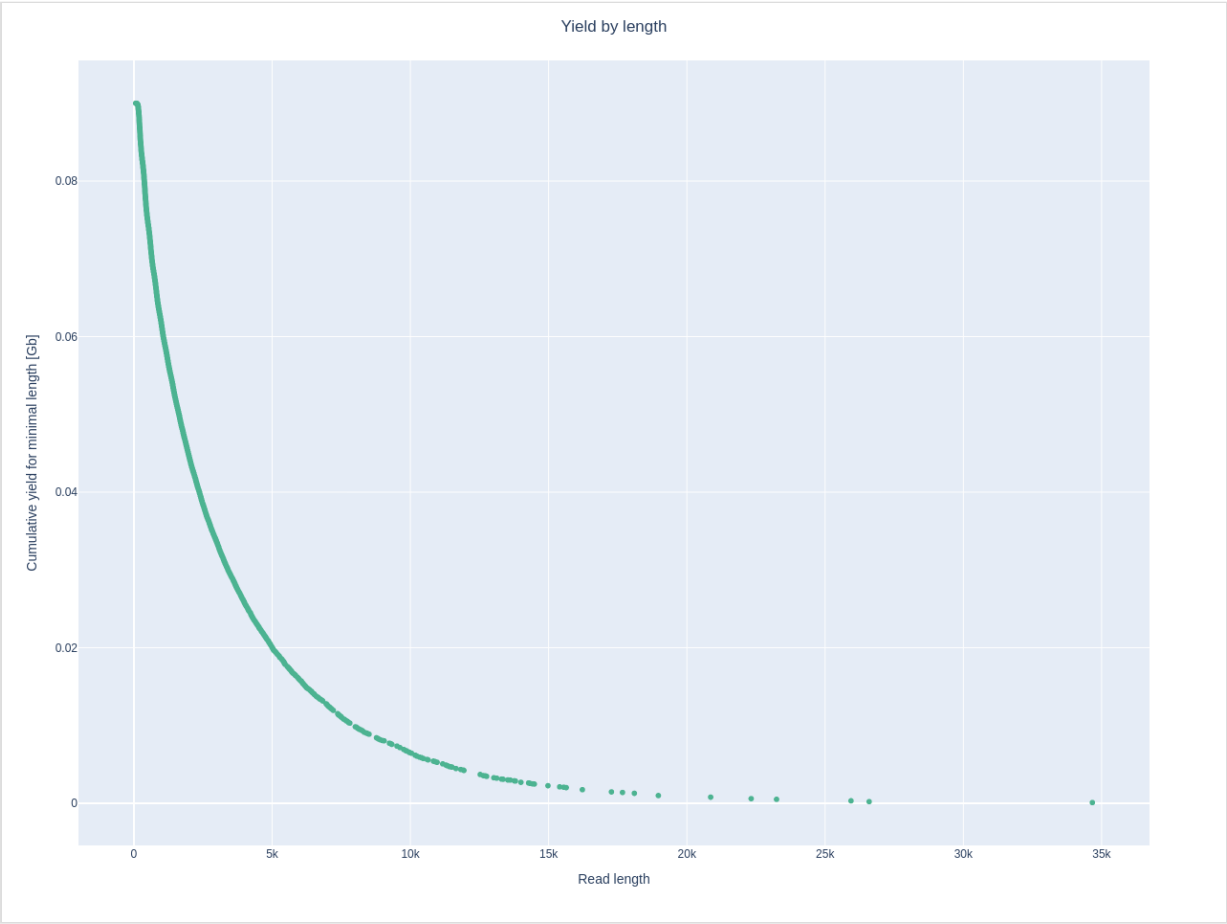
2: 34666 (20.0)

3: 33788 (19.2)

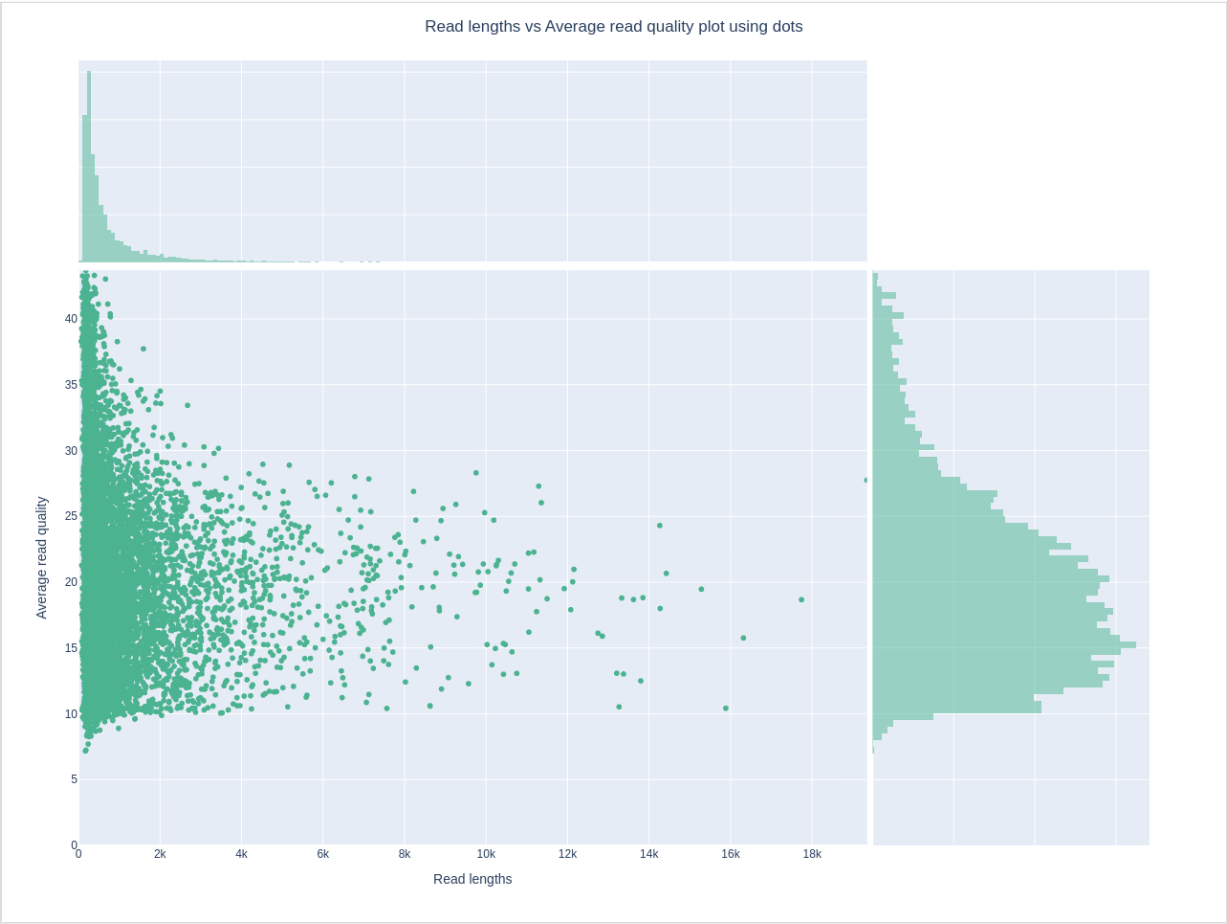
4: 29264 (20.9)

5: 28957 (19.7)

Yield by Length



Length vs Quality



Post-Filtration QC

Statistics

General summary:

Mean read length:	2,562.2
Mean read quality:	16.8
Median read length:	1,759.0
Median read quality:	19.0
Number of reads:	24,804.0
Read length N50:	3,192.0
STDEV read length:	2,333.7
Total bases:	63,553,941.0

Number, percentage and megabases of reads above quality cutoffs

>Q10:	24740	(99.7%)	63.5Mb
>Q15:	19293	(77.8%)	50.0Mb
>Q20:	10680	(43.1%)	27.2Mb
>Q25:	3210	(12.9%)	7.0Mb
>Q30:	581	(2.3%)	0.9Mb

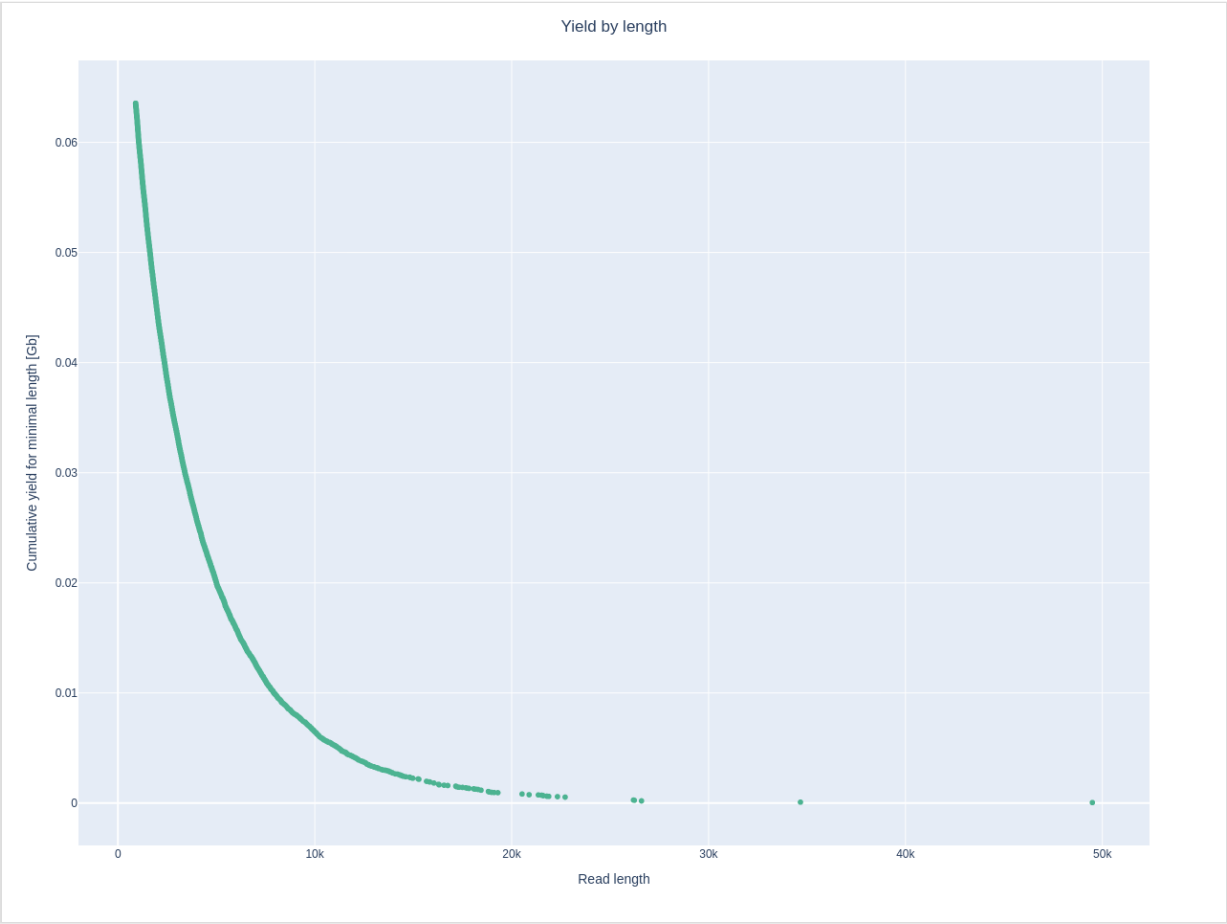
Top 5 highest mean basecall quality scores and their read lengths

1:	43.9	(914)
2:	42.6	(1087)
3:	41.6	(1038)
4:	40.5	(1039)
5:	39.9	(967)

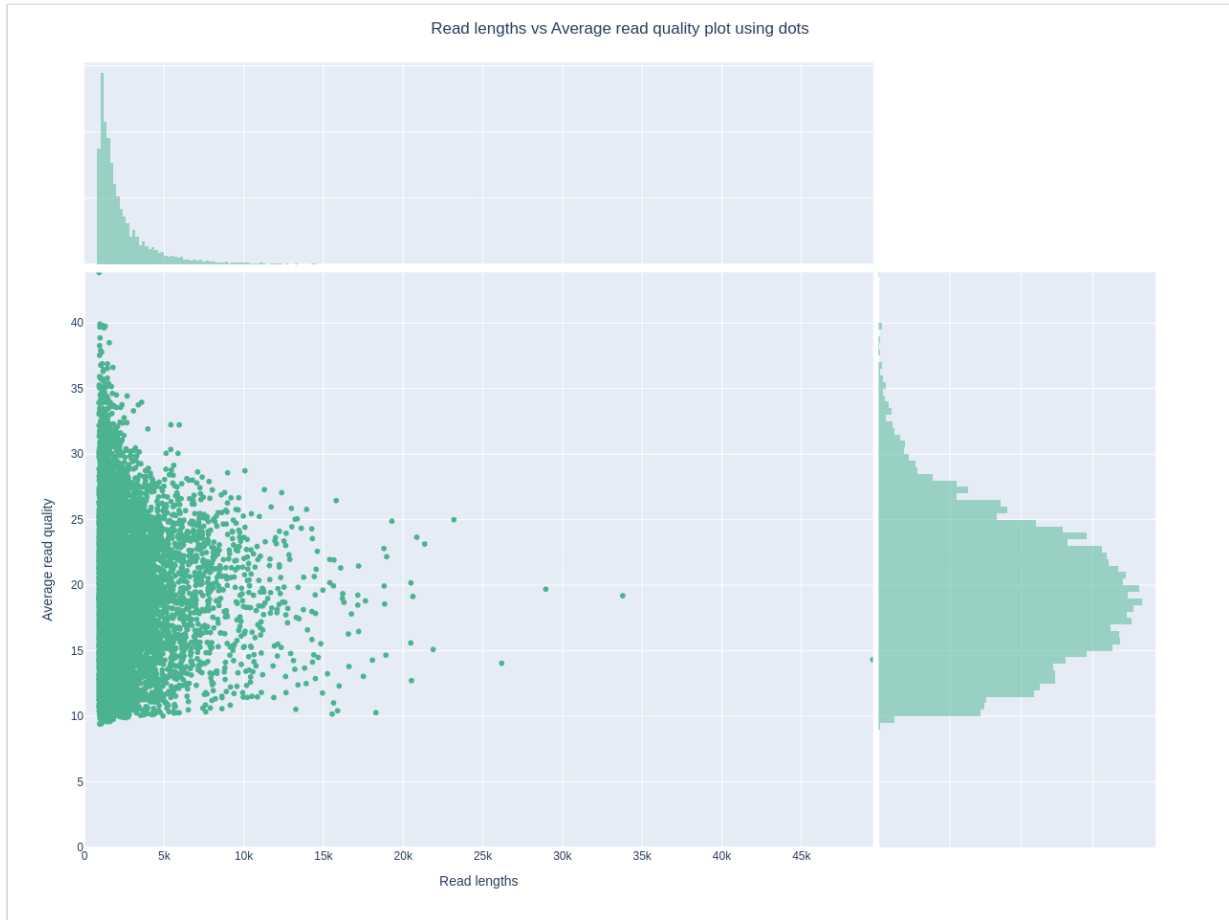
Top 5 longest reads and their mean basecall quality score

1:	49492	(14.3)
2:	34666	(20.0)
3:	33788	(19.2)
4:	29264	(20.9)
5:	28957	(19.7)

Yield by Length



Length vs Quality



Method:

QC pipeline steps:

1. Pre-QC Analysis: Raw reads analyzed with NanoPlot
2. Read Filtering: Reads filtered with FiltLong (min length: 900bp, min mean Q: 15)
3. Post-QC Analysis: Filtered reads analyzed with NanoPlot

Analysis:

Sorprendentemente el agua tuvo el mejor rendimiento, con mas Mb y la mejor calidad los parametros tal vez sean un poco muy restrictivos para T1 a pesar de que creo que son ideales para el resto de las muestras. A pesar de lo que pensabamos T1 tuvo el peor rendimiento, tal vez el tejido de coral tiene muchos inhibidores que impidieron el secuenciamiento ademas que muchos hay muchos fragmentos pequeños de ~500 pb, pero antes de normalizar T1 fue la muestra con mas ng/uL de las presentadas, voy a continuar asi a ver que tal.